SHREYANSH RAI

Full Stack Developer | MERN Stack Specialist | AI & ML Engineer

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PROFESSIONAL SUMMARY

Full Stack MERN Developer & AI/ML Engineer with 1+ year of production experience. Built and deployed accudesign.in at Technewity Labs. Spearheaded a published Hybrid Deep Learning research project (BiLSTM + XGBoost + Autoencoder) for UPI fraud detection achieving 99.9% recall. B.E. Computer Science (AI & ML), ARMIET Mumbai | CGPA: 8.63.

TECHNICAL SKILLS

Frontend: React.js, HTML5, CSS3, JavaScript (ES6+), Hooks, Context API, Redux, Responsive Design
Backend: Node.js, Express.js, RESTful APIs, JWT Auth, OAuth 2.0, FastAPI (Python), MVC Architecture
AI / ML: TensorFlow/Keras, Scikit-learn, XGBoost, BiLSTM, Autoencoders, Isolation Forest, SMOTE, Pandas, NumPy, Streamlit
Database: MongoDB, Mongoose ODM, Schema Design, Aggregation Pipelines, Indexing & Query Optimization, My SQL, SQL
DevOps / Tools: Git/GitHub, Postman, Render, Vercel, VS Code, Agile/Scrum, Arduino IDE, ESP32

WORK EXPERIENCE

Full Stack Developer Intern · AI & ML Engineer · Technewity Labs, Pune *Jan 2025 – Apr 2026*

- Built & deployed ACCU DESIGN (accudesign.in) production-grade MERN platform with role-based access (Admin/User/Vendor), real-time order tracking, and admin dashboards for enterprise clients.
- Designed 15+ RESTful API endpoints with JWT auth, middleware validation & role-based access control; optimized MongoDB schemas for 40%+ query performance gain.
- Integrated AI chatbot API and applied ML model insights into web platform; bridged full-stack and AI/ML workflows in a production environment.

Web Development Intern · Codespaze, Mumbai *Jul 2024 – Sep 2024*

- Enhanced frontend performance across client web apps (HTML5, CSS3, JS); supported REST API integrations using Git-based Agile workflows.

RESEARCH & PUBLICATION

Precise and Enhanced Identification of UPI-based Transaction Scams using Hybrid Ensemble Model — *ACROSET 2026 Conference (IEEE)*

- Lead Author - Architected a V5 Hybrid Deep Learning system (BiLSTM + Bahdanau Attention + XGBoost + Random Forest + Isolation Forest + Autoencoder) to detect UPI transaction fraud in real time.
- Achieved 99.8% Recall, 99.8% Precision, and 99.9% AUC-ROC on the 6.36M-record PaySim dataset; system processes 10,000+ transactions/second via FastAPI asynchronous backend.
- Deployed an interactive Streamlit dashboard with real-time monitoring, explainable AI alerts, and a built-in transaction simulator for zero-day attack testing.

KEY PROJECTS

ACCU DESIGN live at accudesign.in | *React.js · Node.js · Express.js · MongoDB · JWT · Vercel / Render* *Oct 2024 – Present*

- Production-deployed service platform with multi-role dashboards, complete order lifecycle management, JWT + OAuth security, and 15+ REST API endpoints.
- Separated concerns across distinct service layers (Auth Service, Order Service, User Service) laying groundwork for future microservices migration.

UPI Fraud Detection live at netra-fraud-dashboard | *Python · TensorFlow · XGBoost · Random Forest · BiLSTM · FastAPI* *2025*

- Dual-path hybrid AI engine: Path A (XGBoost + RF) auto-blocks known fraud; Path B (BiLSTM + Autoencoder + Isolation Forest) detects novel zero-day attacks via temporal sequence analysis.
- Engineered velocity & behavioural features from PaySim dataset; applied SMOTE + Z-score normalization on 6.36M records; 99.7% accuracy, 99.88% F1-Score.

EDUCATION

B.E. Computer Science (AI & ML) — ARMIET, Mumbai University *Graduating 2026 | CGPA: 8.63*
HSC (ISC Board) — St. Joseph Montessori School, Lucknow *2021 | 89.27%*

CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Web Development Internship — Technewity Labs, Mumbai (Dec 2025)
- Python 3.X Programming — Simplilearn SkillUp (Feb 2024)
- Mumbai Hacks 2025 — Largest Agentic AI Hackathon participant (Nov 2025)